

Data Visualization Mini Project : 120 Years of Olympic History

Introduction

The dataset used for this analysis is the "Olympic Games 1896-2016" dataset (`athlete_events.csv`). It contains historical data on athletes, including their physical attributes, nationality, sport, and medal achievements.

The objective of the analysis is to clean ,aggregate and visualize historical Olympic data to discover trends in Country participation, gender distribution, physical attributes of athletes correlations between age, height and weight.

Dataset Description

Source - Open source Olympics historical dataset.

Important Variables Used - Name , Sex , Age , Height , Weight , NOC (Country Code), Year , Season , Sport , and Medal .

```
In [1]: # Importing necessary libraries for data manipulation and visualization
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Set a clean visual style for all our plots and a cleaner output
sns.set_theme(style="whitegrid")
import warnings
warnings.filterwarnings('ignore')
```

Data Processing

In this step we will :

1. Load the dataset.
2. Handle missing values (e.g., replacing NaN in the 'Medal' column with 'No Medal').
3. Filter the data to focus exclusively on the **Summer Olympics** to make our historical trends consistent.

```
In [2]: df = pd.read_csv('athlete_events.csv')

# Handling missing values
# If an athlete didn't win a medal, the value is NaN. We fill it with 'No Medal'
df['Medal'] = df['Medal'].fillna('No Medal')

# Data Cleaning
# Drop exact duplicate rows if any exist
df = df.drop_duplicates()

# Filtering
# We filter the dataset to only analyze the Summer Olympics
summer_df = df[df['Season'] == 'Summer']

# Display the first 10 rows to confirm preprocessing
summer_df.head(10)
```

Out[2]:

	ID	Name	Sex	Age	Height	Weight		Team	NOC	Games	Year
0	1	A Dijiang	M	24.0	180.0	80.0		China	CHN	1992 Summer	1992
1	2	A Lamusi	M	23.0	170.0	60.0		China	CHN	2012 Summer	2012
2	3	Gunnar Nielsen Aaby	M	24.0	NaN	NaN		Denmark	DEN	1920 Summer	1920
3	4	Edgar Lindenau Aabye	M	34.0	NaN	NaN	Denmark/Sweden	DEN		1900 Summer	1900
26	8	Cornelia "Cor" Aalten (- Strannood)	F	18.0	168.0	NaN		Netherlands	NED	1932 Summer	1932
27	8	Cornelia "Cor" Aalten (- Strannood)	F	18.0	168.0	NaN		Netherlands	NED	1932 Summer	1932
29	10	Einar Ferdinand "Einari" Aalto	M	26.0	NaN	NaN		Finland	FIN	1952 Summer	1952
31	12	Jyri Tapani Aalto	M	31.0	172.0	70.0		Finland	FIN	2000 Summer	2000
32	13	Minna Maarit Aalto	F	30.0	159.0	55.5		Finland	FIN	1996 Summer	1996
33	13	Minna Maarit Aalto	F	34.0	159.0	55.5		Finland	FIN	2000 Summer	2000

Visualization and Analysis

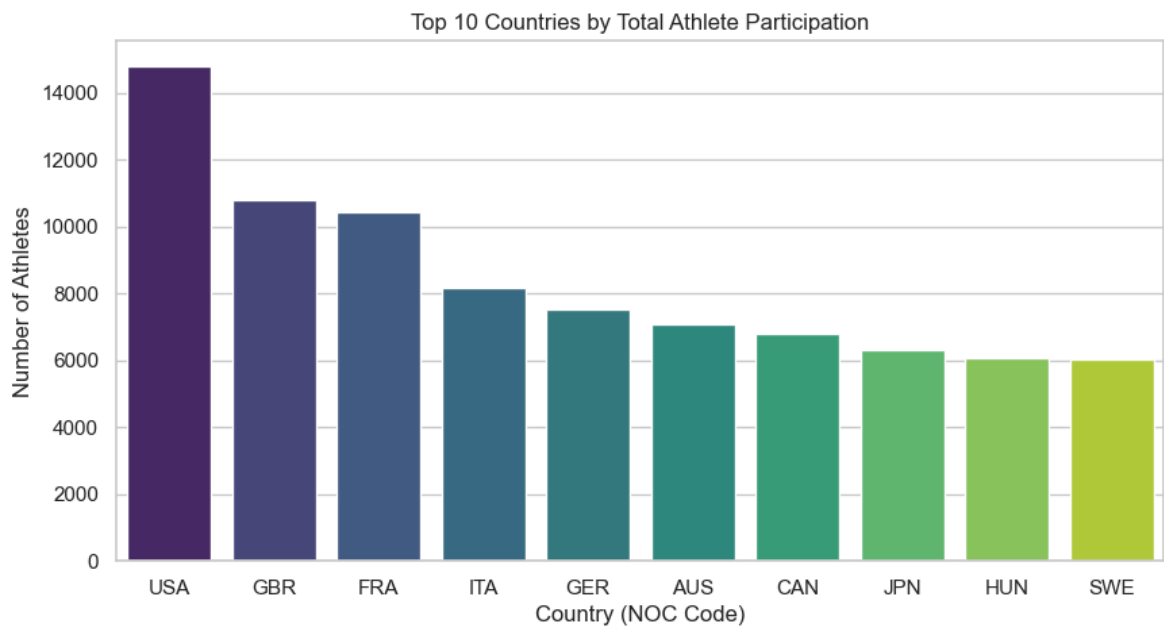
Plot 1: Top 10 Participating Countries (Bar Chart)

We aggregate the total number of athletes sent by each country over the years.

In [3]:

```
plt.figure(figsize=(10,5))
# Grouping by Country Code (NOC) and taking the top 10
top_countries = summer_df['NOC'].value_counts().head(10)
```

```
sns.barplot(x=top_countries.index, y=top_countries.values, palette="viridis")
plt.title('Top 10 Countries by Total Athlete Participation')
plt.xlabel('Country (NOC Code)')
plt.ylabel('Number of Athletes')
plt.show()
```



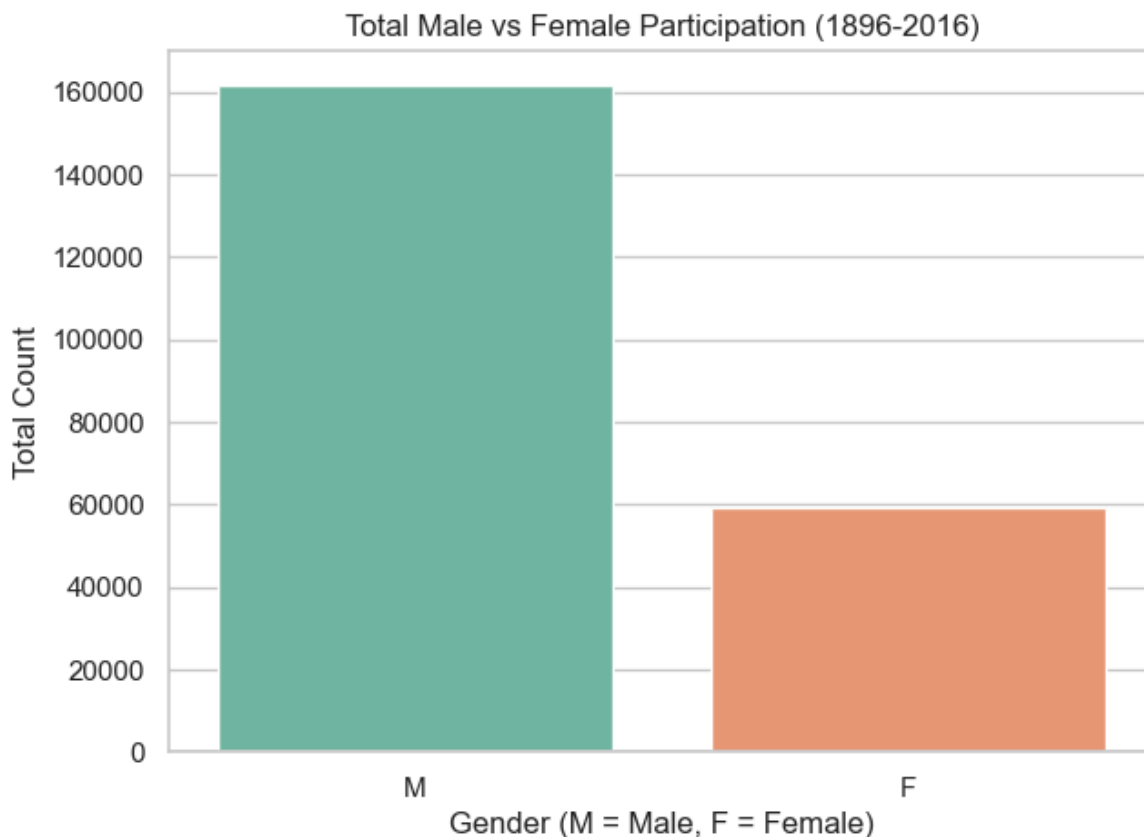
Insight:

The United States (USA), France (FRA), and Great Britain (GBR) have historically sent the most athletes to the Summer Olympics.

Plot 2: Gender Distribution (Bar Chart)

Let's look at the overall comparison of male vs. female participants.

```
In [4]: plt.figure(figsize=(7,5))
sns.countplot(x='Sex', data=summer_df, palette='Set2')
plt.title('Total Male vs Female Participation (1896-2016)')
plt.xlabel('Gender (M = Male, F = Female)')
plt.ylabel('Total Count')
plt.show()
```



Insight:

Historically, there have been significantly more male athletes than female athletes overall across the 120 years.

Plot 3 and 4 : Historical Trends (Line Charts)

How has the scale of the Olympics grown over time in terms of athletes and participating nations?

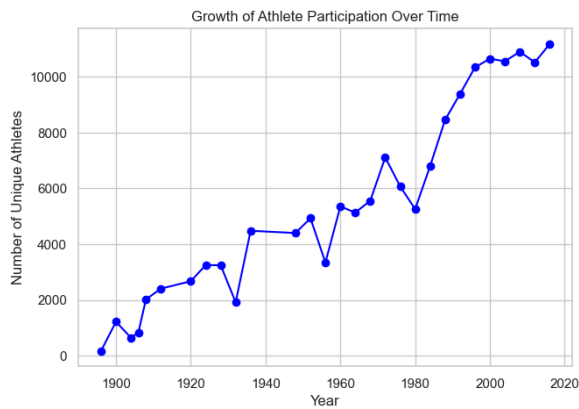
```
In [5]: # Grouping data by Year to find unique athletes and unique nations
athletes_over_time = summer_df.groupby('Year')['ID'].nunique()
nations_over_time = summer_df.groupby('Year')['NOC'].nunique()

# Creating a figure with 2 subplots side by side
fig, ax = plt.subplots(1, 2, figsize=(16, 5))

# Plot 3: Athletes over time
ax[0].plot(athletes_over_time.index, athletes_over_time.values, marker='o', color='red')
ax[0].set_title('Growth of Athlete Participation Over Time')
ax[0].set_xlabel('Year')
ax[0].set_ylabel('Number of Unique Athletes')

# Plot 4: Nations over time
ax[1].plot(nations_over_time.index, nations_over_time.values, marker='s', color='blue')
ax[1].set_title('Growth of Participating Nations Over Time')
ax[1].set_xlabel('Year')
ax[1].set_ylabel('Number of Nations')

plt.show()
```



Insight:

Both the number of athletes and participating nations show an upward trend, proving the Olympics have become a massive global event. The sharp drops correspond to the World Wars.

Plot 5 and 6 : Physical Attributes (Histograms)

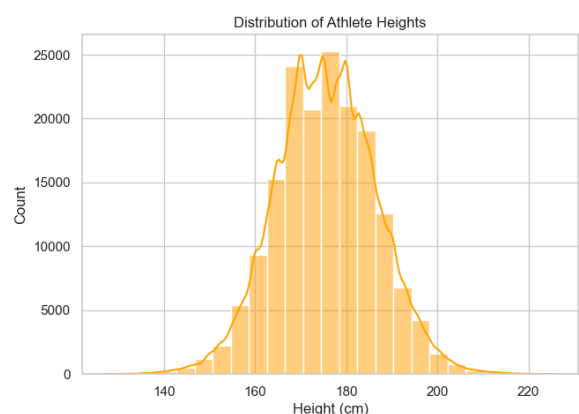
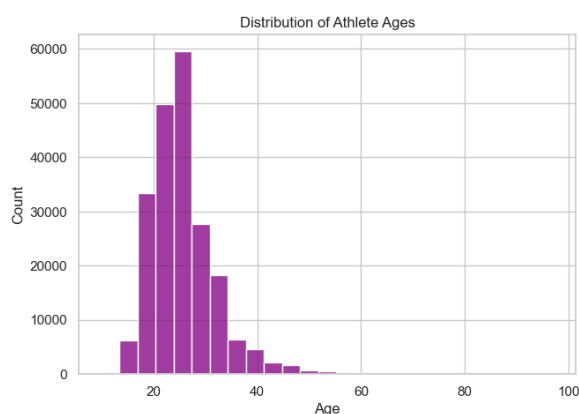
Observing the distribution of the athletes' ages and heights.

```
In [6]: fig, ax = plt.subplots(1, 2, figsize=(16, 5))

# Plot 5: Age Distribution
sns.histplot(summer_df['Age'].dropna(), bins=25, color='purple', kde=False, ax=ax[0])
ax[0].set_title('Distribution of Athlete Ages')
ax[0].set_xlabel('Age')

# Plot 6: Height Distribution
sns.histplot(summer_df['Height'].dropna(), bins=25, color='orange', kde=True, ax=ax[1])
ax[1].set_title('Distribution of Athlete Heights')
ax[1].set_xlabel('Height (cm)')

plt.show()
```



Insight :

Most Olympic athletes are in their early-to-mid 20s. The height distribution forms a nearly perfect normal distribution (bell curve) centered around 175 cm.

Plot 7 & 8: Height vs. Weight (Scatter Plots)

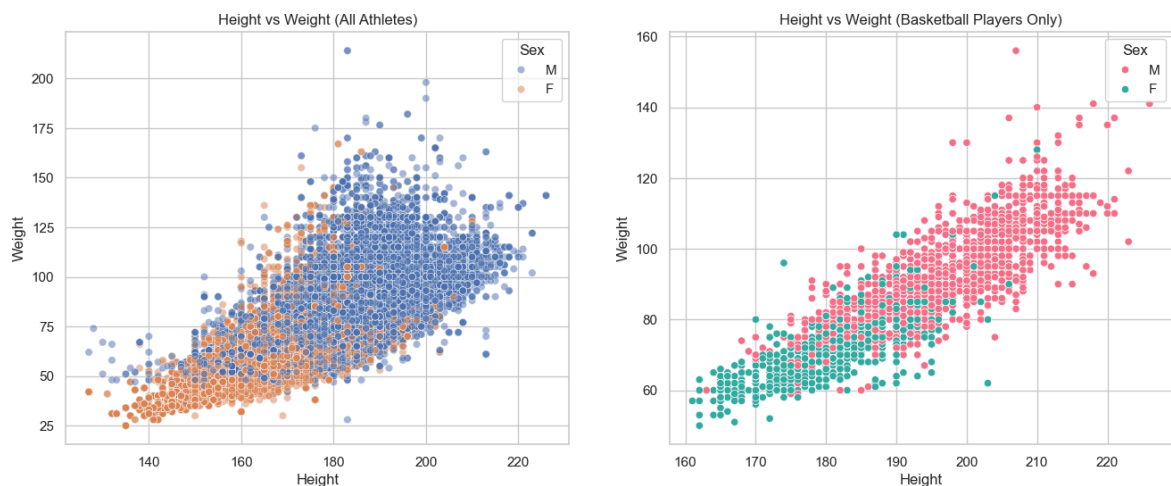
How do height and weight correlate, and how does it differ in specific sports like Basketball?

```
In [11]: fig, ax = plt.subplots(1, 2, figsize=(16, 6))

# Plot 7: General Height vs Weight colored by Sex
sns.scatterplot(x='Height', y='Weight', data=summer_df, hue='Sex', alpha=0.5, ax=ax[0]).set_title('Height vs Weight (All Athletes)')

# Plot 8: Height vs Weight for Basketball only
basketball_df = summer_df[summer_df['Sport'] == 'Basketball']
sns.scatterplot(x='Height', y='Weight', data=basketball_df, hue='Sex', palette='
ax[1].set_title('Height vs Weight (Basketball Players Only)')

plt.show()
```



Insight :

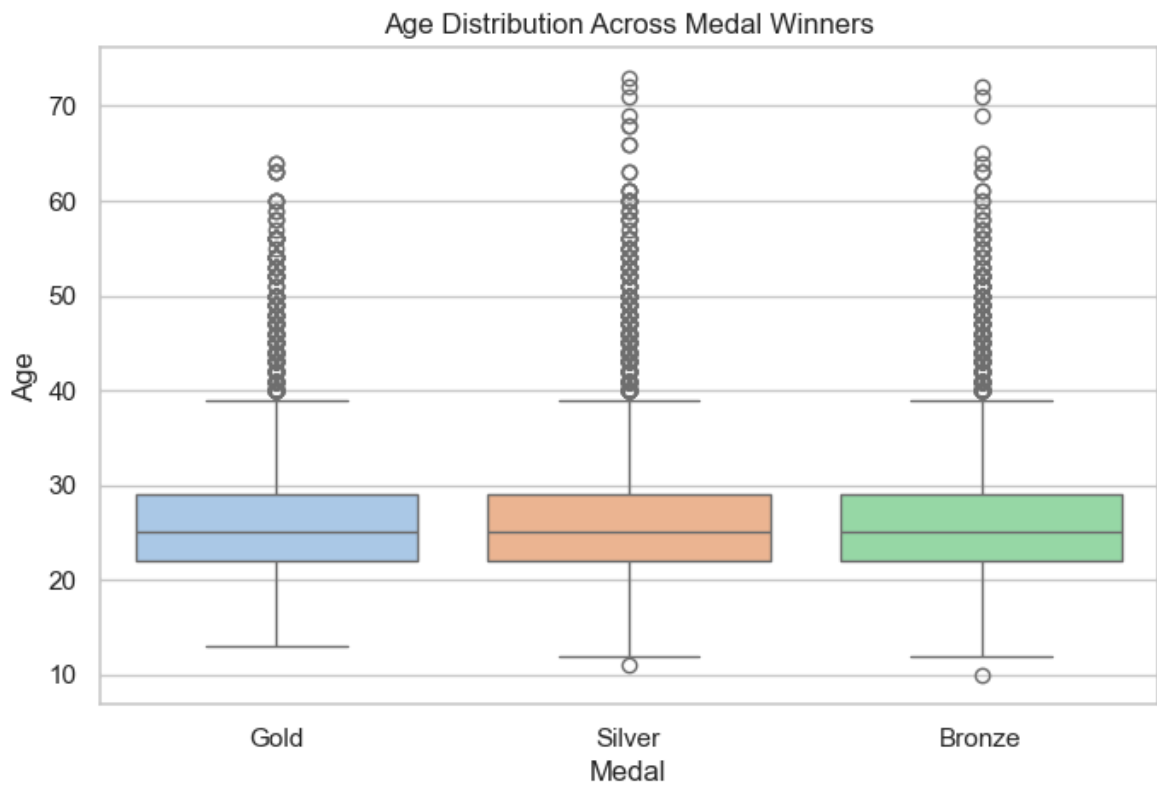
There is a clear positive correlation between height and weight. Males generally cluster higher on both axes. Basketball players are distinctly taller and heavier than the average athlete.

Plot 9 : Age across Medal Winners (Box Plot)

Does age vary depending on the type of medal won?

```
In [8]: plt.figure(figsize=(8,5))
# Filter out non-medalists for this plot
medal_winners = summer_df[summer_df['Medal'] != 'No Medal']

sns.boxplot(x='Medal', y='Age', data=medal_winners, order=['Gold', 'Silver', 'Br
plt.title('Age Distribution Across Medal Winners')
plt.show()
```



Insight :

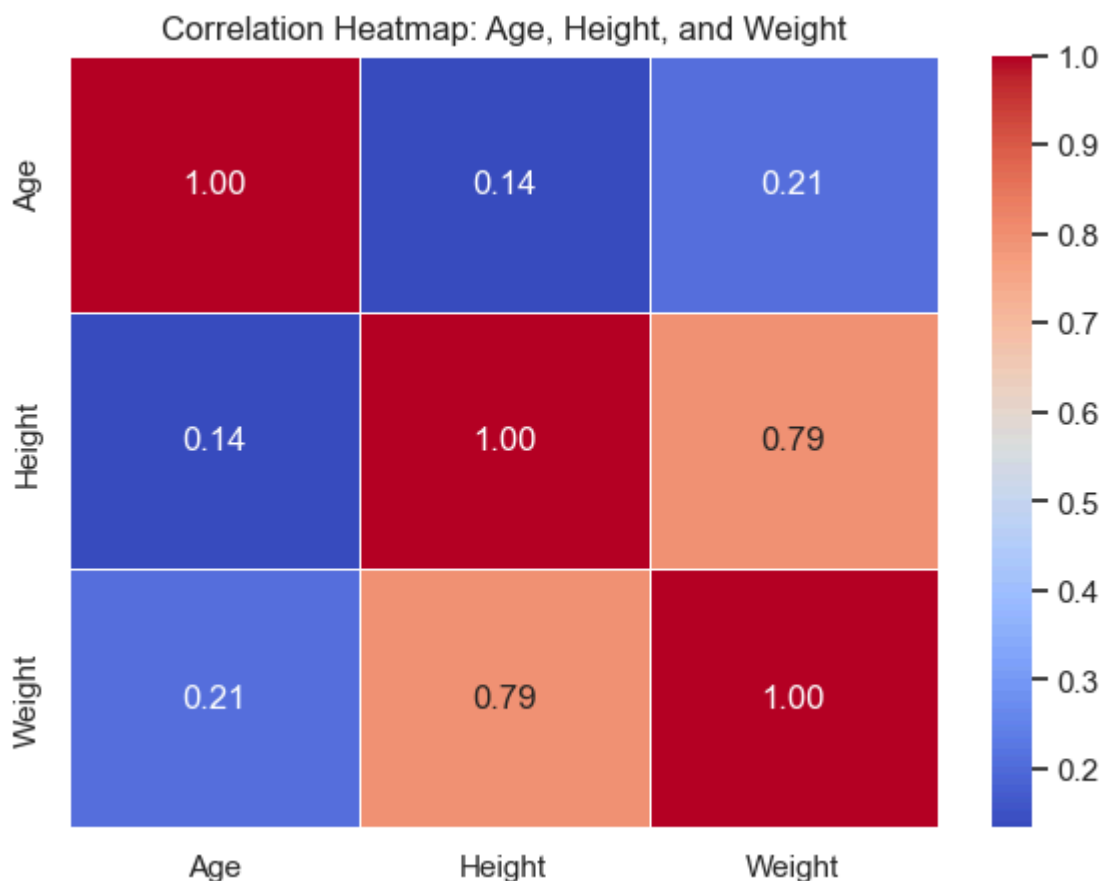
The median age for Gold, Silver, and Bronze medalists is virtually identical (around 24-25 years old). However, there are older outliers (athletes in their 50s and 60s) winning medals, likely in sports like Equestrian or Shooting.

Plot 10 : Attribute Correlation (Heatmap)

A mathematical look at how Age, Height, and Weight are related.

```
In [9]: plt.figure(figsize=(7,5))
# Select only numeric columns for correlation
correlation_data = summer_df[['Age', 'Height', 'Weight']].corr()

sns.heatmap(correlation_data, annot=True, cmap='coolwarm', fmt=".2f", linewidths=1)
plt.title('Correlation Heatmap: Age, Height, and Weight')
plt.show()
```



Insight :

Height and Weight are highly positively correlated (0.80), which makes physical sense. Age has a very weak correlation with height and weight, meaning physical size does not strictly increase with age for adult Olympians."

Conclusion

Summary of the Analysis

This project successfully analyzed 120 years of Olympic history. Through data wrangling and missing-value handling, we cleaned a massive dataset and extracted focused insights regarding the Summer games.

Insights gained from visualizations

1. The Olympic games have seen massive, steady growth in both global participation and the number of competing athletes.
2. The United States dominates historically in terms of sheer participation volume.
3. While males have historically dominated participation numbers, visualizing the data shows clear clusters of physical attributes (Height vs. Weight) that separate sports (like Basketball) from the general average.
4. Medal-winning is highly concentrated in athletes aged 22-26, irrespective of the medal color.

Tools Used

Language : Python 3 Libraries : Pandas (Data manipulation), Matplotlib & Seaborn (Data Visualization), Numpy (Calculations). Environment : Jupyter Notebook

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Section - CD

In []: